

MUHAMMAD MUBASHAR

Computer Systems Engineering Undergraduate · IoT & Embedded Systems

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[LinkedIn](#) · [Portfolio](#) · [GitHub](#)

SUMMARY

Computer Systems Engineering undergraduate at Mehran University of Engineering & Technology with a sustained focus on embedded systems, IoT, and the boundary between hardware and software. Confident programming in C/C++, Java, Python, and SQL; comfortable prototyping with Arduino, ESP32, and Raspberry Pi Pico. Currently deepening expertise in LoRaWAN networking and MQTT-based IoT communication to build stable, scalable, and automated solutions.

EDUCATION

Bachelor of Engineering, Computer Systems

Aug 2024 – May 2028 (Expected)

Mehran University of Engineering & Technology (MUET)

Hyderabad, Pakistan

Relevant coursework: Data Structures & Algorithms, Object-Oriented Programming, Operating Systems (Scheduling & Memory Management), Database Management Systems, Digital Electronics, Signals & Systems, Discrete Mathematics.

TECHNICAL SKILLS

Languages: C, C++, Java, Python, SQL, MATLAB

Embedded & IoT: Arduino Uno, ESP32, ESP8266, Raspberry Pi Pico, Sensor Integration, Relay Control, LoRaWAN (in progress)

Software & CS: OOP, DSA, Database Design (1NF–3NF), ACID Transactions, OS Concepts, Signals & Systems

Tools & Cloud: Arduino IDE, MySQL, PHP (PDO), HTML5/CSS3, Git/GitHub, Oracle Cloud Infrastructure

SELECTED PROJECTS

Lab Environmental Monitoring System

Apr 2025 – May 2025

Arduino Uno · C/C++ · BH1750 · DHT11

- Engineered a real-time hardware-software solution to monitor ambient light intensity and temperature in chemical- and medical-test environments.
- Achieved high-precision lux readings using the BH1750 digital sensor and modular firmware prepared for IoT extension.

MediCare — Hospital Management Database

Academic Project

MySQL (InnoDB) · PHP (PDO) · HTML5/CSS3

- Designed an enterprise-grade relational schema normalized to 3NF, eliminating insertion, update, and deletion anomalies.
- Enforced referential integrity through primary keys, ENUMs, and cascaded deletes; implemented ACID transactions with PDO `beginTransaction` / `commit` / `rollback`.

Automated Water Overflow Control System

Intermediate Project

Arduino · C/C++ · Ultrasonic Sensor · Relay Module

- Built a closed-loop microcontroller system performing continuous ultrasonic depth measurement to autonomously prevent tank overflow.
- Integrated low/high threshold logic with a relay module to control a high-voltage pump, reducing water wastage through simple automation.

CERTIFICATIONS

Oracle Cloud Infrastructure 2025 — Certified AI Foundations Associate

Oracle

Professional Certification in Python Programming

Athena Global Education (UniAthena)

Cambridge International Qualification

Cambridge Assessment International Education